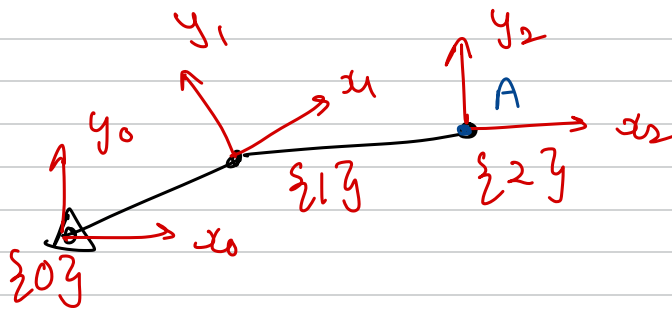


29/9 Tue Lab Task

- ① Pen and Paper
— 2 DOF spatial manipulator FK
 - ② Py script for 6 DOF HEAL and 7 DOF FRANKA FK (using DH parameters).
 - ③ Mujoco FK for Heal and Franka.
- give $\bar{q}$ for Py and mujoco and compare end effector position.

Task 1 Pen and Paper

2 DOF Spatial Manipulator



→ each joint i (frame i) has its own set of $\begin{bmatrix} a_i & d_i & \alpha_i & \theta_i \end{bmatrix} \rightarrow$ DH parameters

$\downarrow$ geometry $\searrow$ Joint parameter (we can choose)

$\therefore$ for each frame, there is an A_i such that

$$P_{i+1} = A_i P_i \Rightarrow A_i = T_i^{i+1} = H_i^{i+1}$$

we know using DH parameters:

$$A_i = Rot_{z, d_i} \cdot Trans_{z, a_i} \cdot Trans_{x, \alpha_i} \cdot Rot_{x, \theta_i}$$

$$\therefore A_i = \begin{bmatrix} c d_i & -s d_i & 0 & 0 \\ s d_i & c d_i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\times \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c d_i & -s d_i & 0 \\ 0 & s d_i & c d_i & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\therefore A_i = \begin{bmatrix} c d_i & -s d_i c d_i & s d_i s d_i & a_i c d_i \\ s d_i & c d_i c d_i & -c d_i s d_i & a_i s d_i \\ 0 & s d_i & c d_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

— from textbook
(Spong v. 1)

end effector : A

$$\therefore \text{from image, } \bar{P}_{A_2} = [0, 0, 0, 1]^T$$

(homogeneous description)

$$\therefore \text{By convention, } \bar{P}_{A_1} = A_2 \bar{P}_{A_2}$$

$$\therefore \bar{P}_{A_0} = A_1 \bar{P}_{A_1}$$

$$\therefore \bar{P}_{A_0} = A_1 A_2 \bar{P}_{A_2}$$

↓
coordⁿ of
end-effector in
base frame {0}

→ Assume θ_1, θ_2 to be the input joint Angles

→ Assume $a_1, d_1, a_2, d_2, a_3, d_3$ to be the geometry parameter from frame 1, 2.

$$\therefore \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} c\alpha_1 & -s\alpha_1 c\alpha_2 & s\alpha_1 s\alpha_2 & a_1 c\alpha_1 \\ s\alpha_1 & c\alpha_1 c\alpha_2 & -c\alpha_1 s\alpha_2 & a_1 s\alpha_1 \\ 0 & s\alpha_1 & c\alpha_1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\times \begin{bmatrix} c\alpha_2 & -s\alpha_2 c\alpha_2 & s\alpha_2 s\alpha_2 & a_2 c\alpha_2 \\ s\alpha_2 & c\alpha_2 c\alpha_2 & -c\alpha_2 s\alpha_2 & a_2 s\alpha_2 \\ 0 & s\alpha_2 & c\alpha_2 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Ans

$$\therefore \bar{P}_{A0} = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} a_1 c\alpha_1 + a_2 c\alpha_1 c\alpha_2 - a_2 s\alpha_1 s\alpha_2 c\alpha_2 + d_2 s\alpha_1 s\alpha_2 \\ a_1 s\alpha_1 + a_2 s\alpha_1 c\alpha_2 + a_2 c\alpha_1 s\alpha_2 c\alpha_2 - d_2 c\alpha_1 s\alpha_2 \\ d_1 + a_2 s\alpha_2 s\alpha_2 + d_2 c\alpha_1 \\ 1 \end{bmatrix}$$

→ ①

Ex $\alpha_1, \alpha_2 \rightarrow$ input Angles given

we can compute (x, y, z) for end effector

for 2DOF spatial using ①

(All DH geometry parameters will be known)